

1-DOF and 2-DOF Spring-System Analysis

The useful result is not just a set of frequencies. An ideal vibration model predicts the structure of the response, while real hardware shifts the measured frequencies through damping, distributed spring mass, sensor mass, and coupled motion.

1-DOF insight

The single-spring test gives natural frequency and damping, but the FFT and peak-count methods reveal different views of the same physical response.

2-DOF insight

The two-spring test requires an eigenvalue model because the motion separates into multiple modal frequencies and mode shapes.

Experimental insight

The accelerometer data validates the theory and also exposes where the printed spring assembly is not a perfect lumped-parameter system.

The vibration workflow moves from model to measurement

The analysis connects hardware setup, theoretical frequency calculations, time-domain response, FFT spectra, damping estimates, and theory-versus-experiment interpretation.

1-DOF workflow	
Theory	f_n
Peak counting	f_d
FFT analysis	resonance peak
Damping	δ, ζ

2-DOF workflow	
Matrix model	$[M], [K]$
Eigenproblem	ω_1, ω_2
Mode shapes	ϕ_1, ϕ_2
Comparison	theory vs. FFT

Discussion focus	
Damping effect	$f_d = f_n \sqrt{1 - \zeta^2}$
Spring behavior	continuous part
Sensor mass	added inertia
Assembly effects	real hardware

Arduino-based sensing captures the free vibration response

AN

Arduino Nano

Reads accelerometer data over I2C and streams the time response.

I2C

Sensor wiring

I2C wiring: VCC to 5V, GND, SCL to A5, SDA to A4.

SPR

Printed springs

One modular spring uses $k = 106.6 \text{ N/m}$.

IMU

GY-521 / MPU-6050

Six-axis sensor board aligned with the spring-motion axis.

AD0

Second sensor address

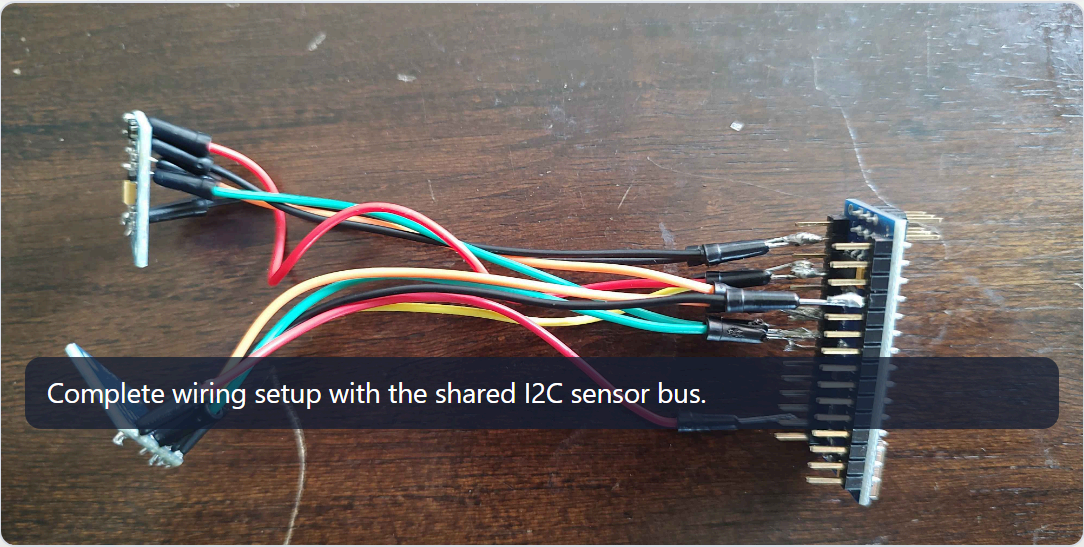
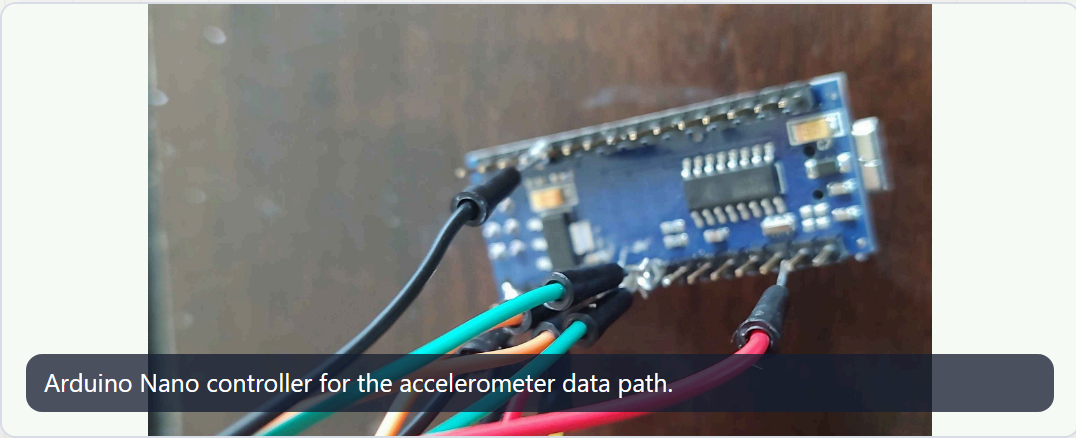
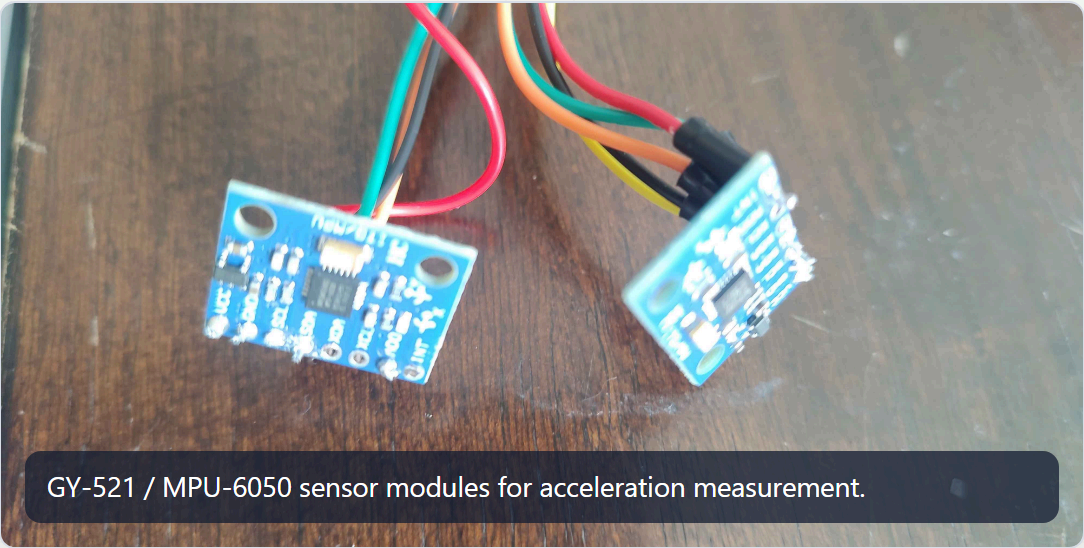
AD0 shifts the second GY-521 from 0x68 to 0x69.

DAQ

Recorded data

Post-processing uses peak count, log decrement, and FFT.

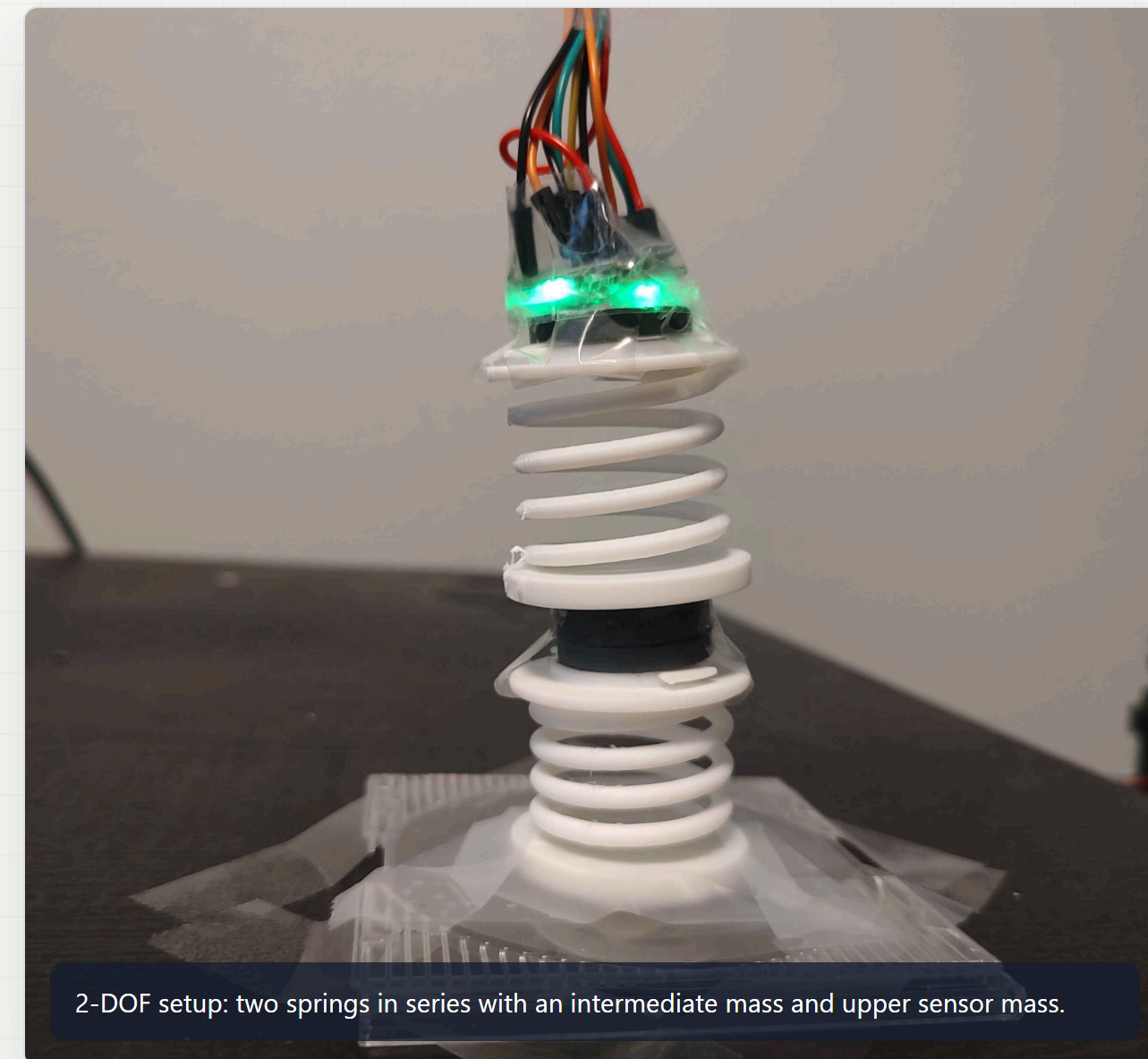
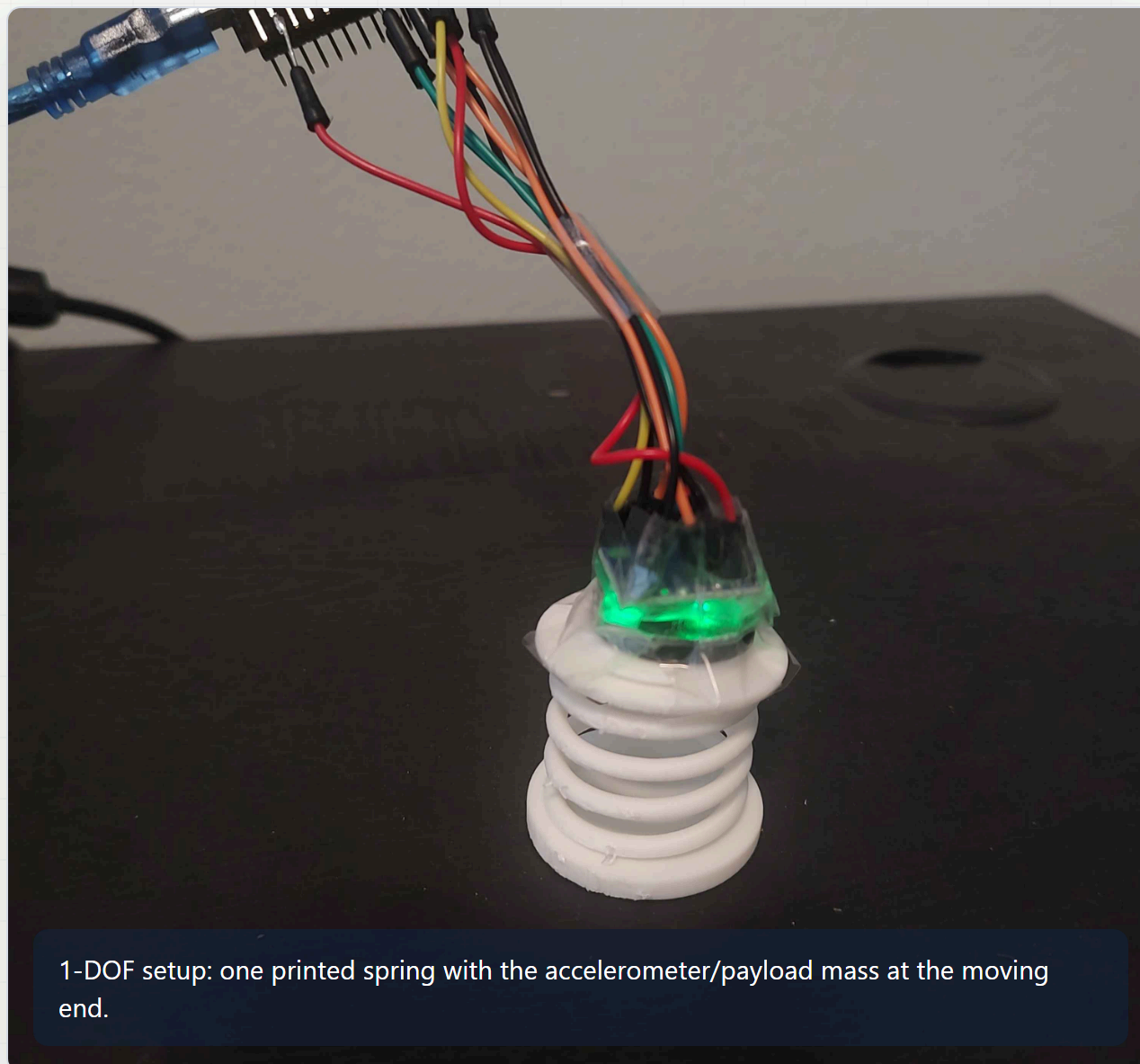
The data path is simple: accelerometers measure motion, the Arduino reads the sensors over I2C, and the recorded acceleration signal is processed into time and frequency results.



SYSTEM SETUP

One spring for 1-DOF, two springs for coupled motion

The physical setups separate the single-coordinate oscillator from the coupled two-coordinate system.



MEASURED TEST VALUES

These values drive the theoretical comparison

The measured test statistics are used consistently in the 1-DOF and 2-DOF calculations.

1-DOF statistics	
m_1	30.4 g
Sampling rate	84.71 Hz
Peak count	11.012 Hz
Log decrement	0.30120
Damping ratio	0.04788
FFT frequency	8.466 Hz

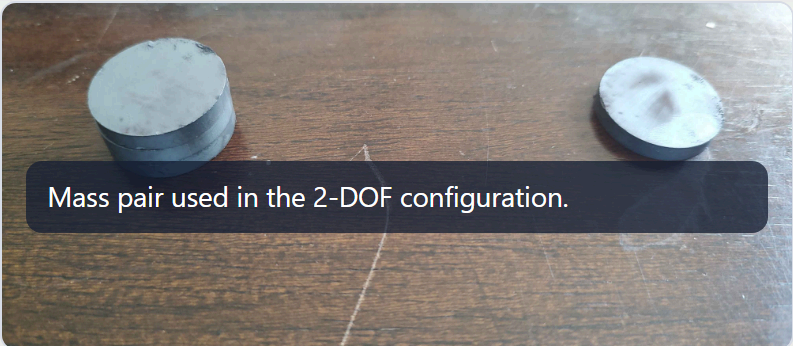
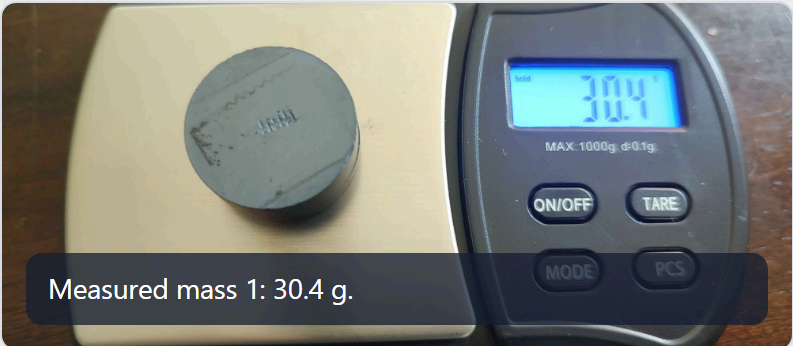
2-DOF statistics	
m_1	30.4 g
m_2	10.1 g
Sampling rate	84.71 Hz
FFT statistic	8.466 Hz
Log decrement	0.30120
Damping ratio	0.04788

Constants used

$$k = 106.6 \text{ N/m}$$

$$m_1 = 0.0304 \text{ kg}, \quad m_2 = 0.0101 \text{ kg}$$

$$f_{\text{peak}} = 11.012 \text{ Hz}, \quad f_{\text{FFT}} = 8.466 \text{ Hz}$$



The free-vibration model predicts frequency and damping

The damping correction is included, but it is small because the measured damping ratio is much less than one.

$$m\ddot{x} + c\dot{x} + kx = 0$$
$$\omega_n = \sqrt{\frac{k}{m}}, \quad f_n = \frac{\omega_n}{2\pi}$$

$$\omega_n = \sqrt{\frac{106.6}{0.0304}} = 59.216 \text{ rad/s}, \quad f_n = \frac{59.216}{2\pi} = 9.425 \text{ Hz}$$

$$\delta = \frac{1}{N} \ln \left(\frac{x_i}{x_{i+N}} \right) = 0.30120, \quad \zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}} = 0.04788$$

$$f_d = f_n \sqrt{1 - \zeta^2} = 9.414 \text{ Hz}$$

1-DOF frequency comparison

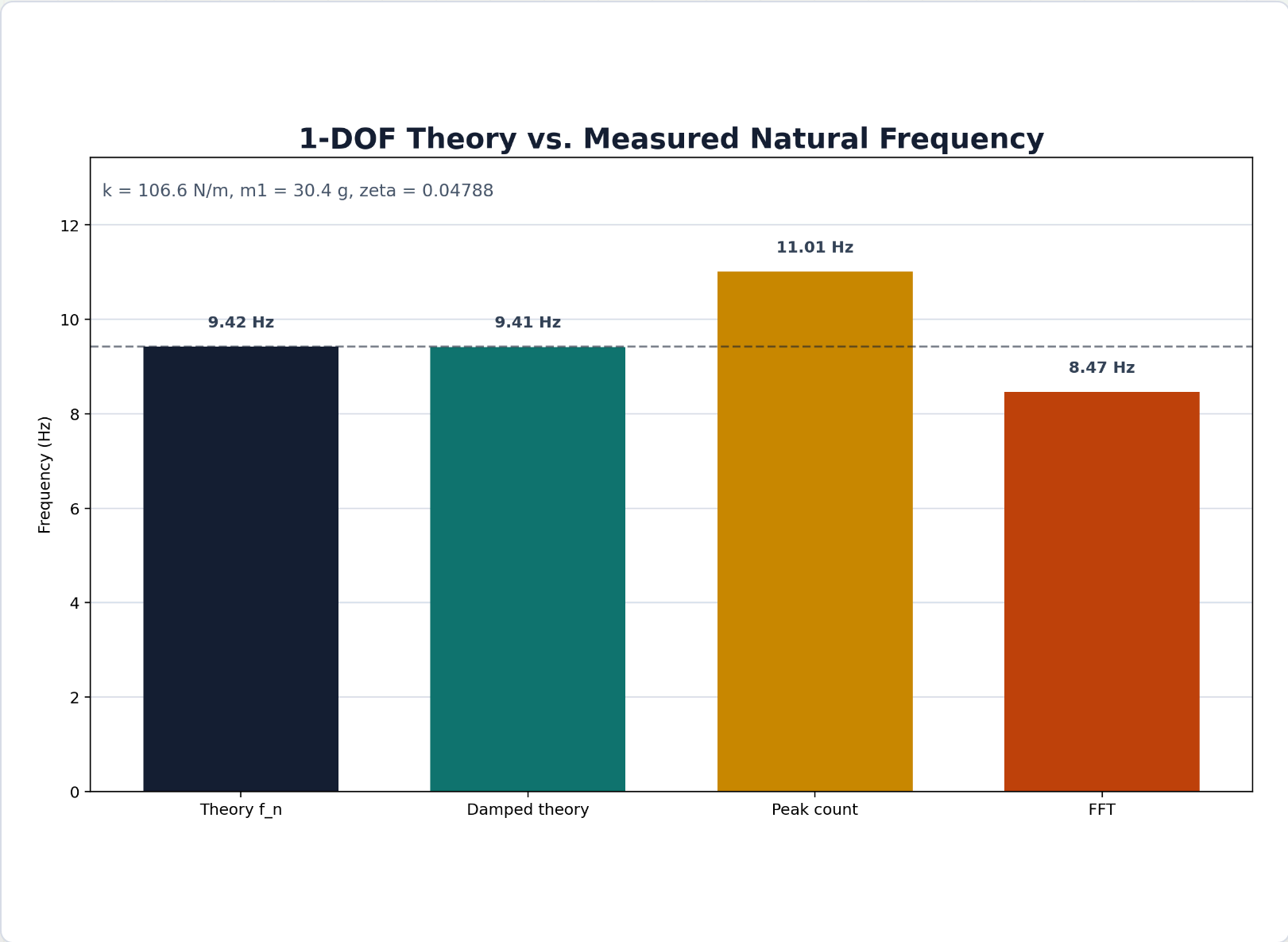
Theoretical ω_n	59.216 rad/s
Theoretical f_n	9.425 Hz
Damped f_d	9.414 Hz
Peak-count result	11.012 Hz
FFT result	8.466 Hz
Damping ratio	0.04788

Because f_d is almost equal to f_n , damping alone does not explain the difference between peak counting and FFT.

1-DOF RESULTS

Peak counting and FFT show different sides of the response

The time-domain method approximates the response as one clean oscillator; the FFT shows the dominant frequency content of the real assembly.



Interpretation

Peak count vs. theory +16.84%

FFT vs. theory -10.17%

Peak count time-domain

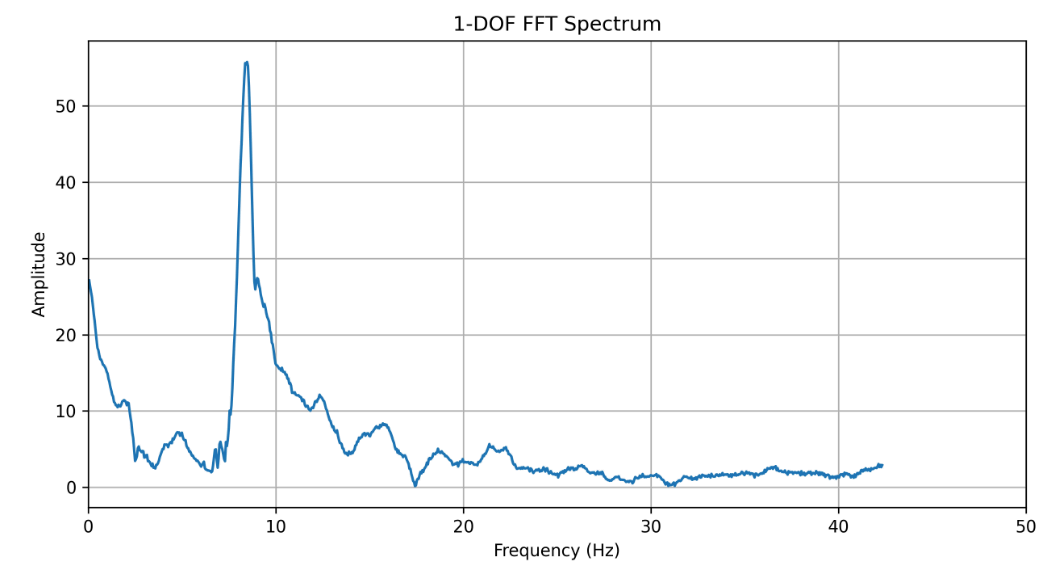
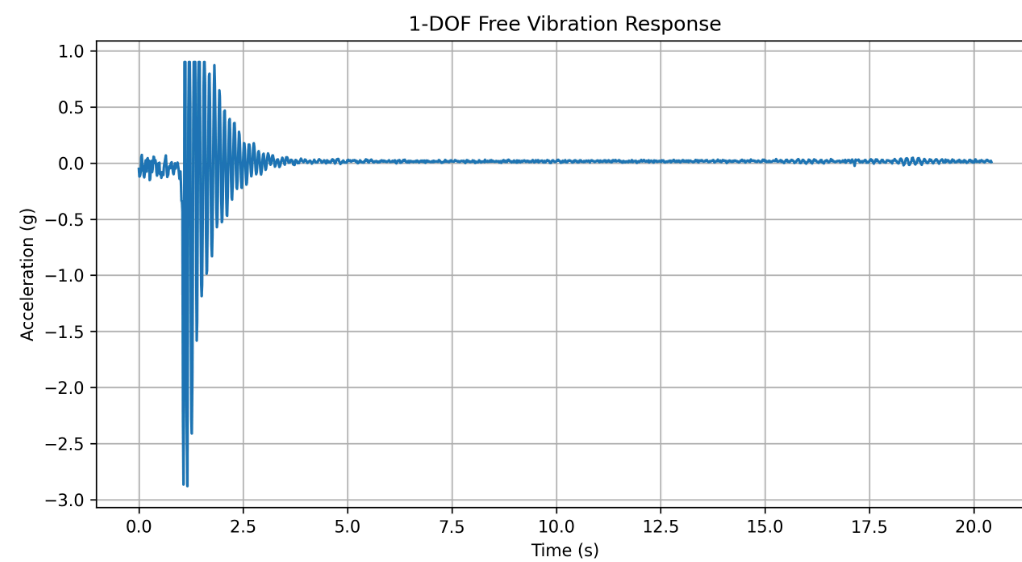
FFT frequency-domain

The disagreement is useful: it shows that the physical printed spring setup is richer than a perfect 1-DOF model.

1-DOF SIGNAL EVIDENCE

The raw time and FFT plots support the measured results

The time-domain response and FFT spectrum answer different questions about the same free-vibration test.



The coupled system is solved as an eigenvalue problem

Two masses and two springs produce two theoretical natural frequencies and two mode shapes.

$$M = \begin{bmatrix} 0.0304 & 0 \\ 0 & 0.0101 \end{bmatrix} \text{ kg}, \quad K = \begin{bmatrix} 213.2 & -106.6 \\ -106.6 & 106.6 \end{bmatrix} \text{ N/m}$$

$$\det(K - \omega^2 M) = \begin{vmatrix} 213.2 - 0.0304\omega^2 & -106.6 \\ -106.6 & 106.6 - 0.0101\omega^2 \end{vmatrix} = 0$$

$$(213.2 - 0.0304\omega^2)(106.6 - 0.0101\omega^2) - (106.6)^2 = 0$$

$$0.00030704\omega^4 - 5.39396\omega^2 + 11363.56 = 0$$

$$\omega_1 = 49.475 \text{ rad/s}, \quad \omega_2 = 122.963 \text{ rad/s}, \quad f_i = \frac{\omega_i}{2\pi}$$

Eigenvalue result

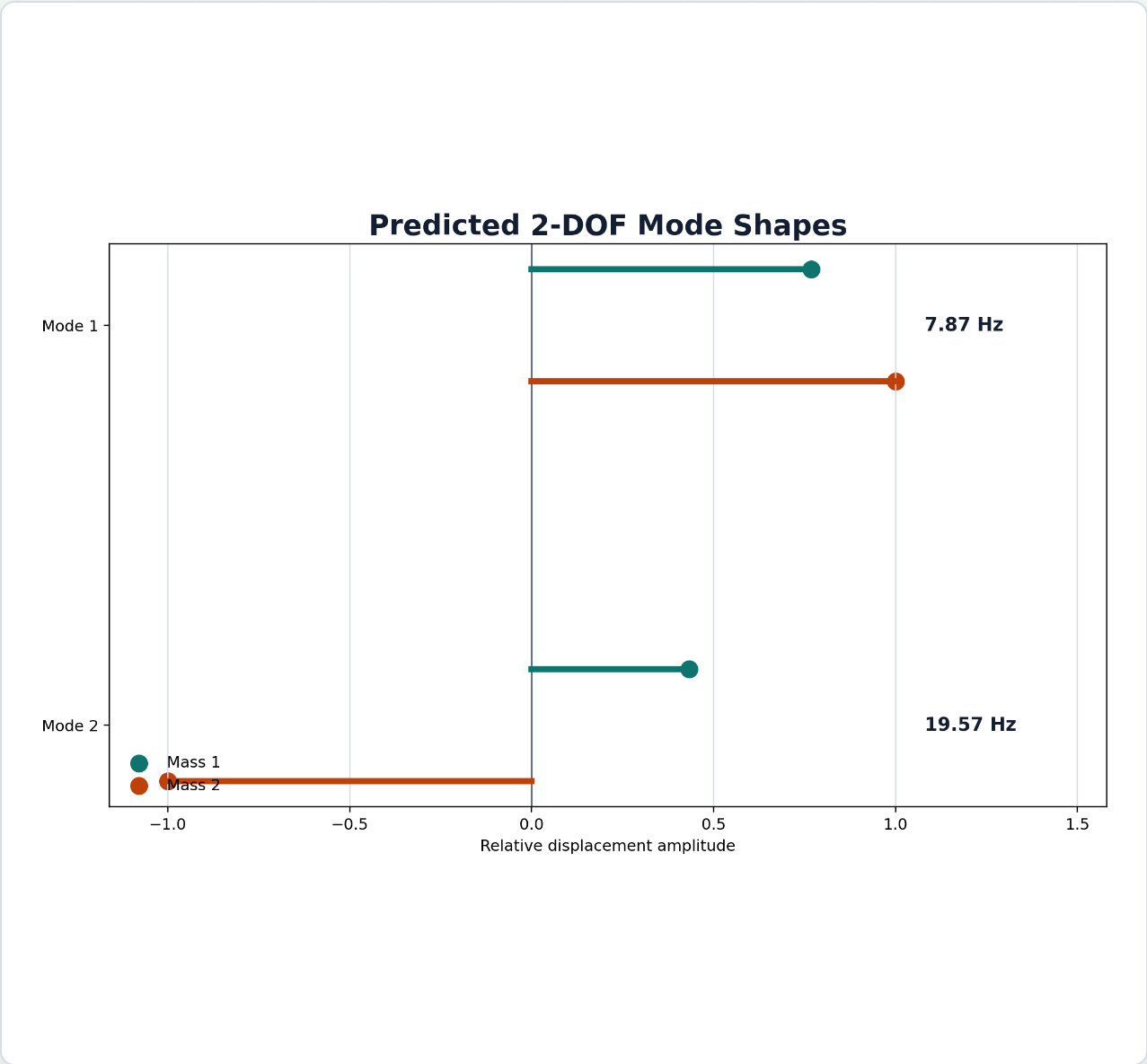
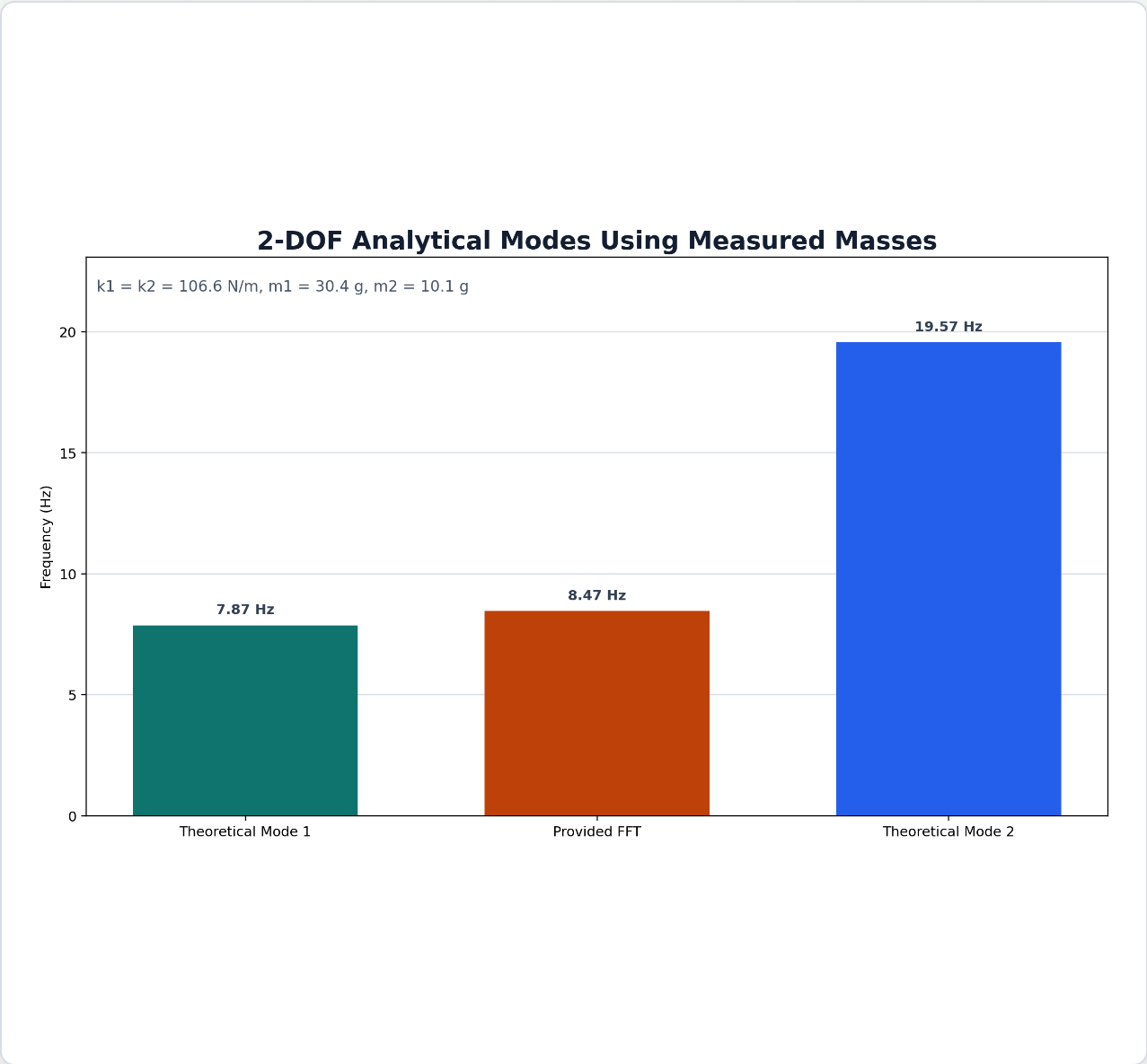
ω_1	49.475 rad/s
ω_2	122.963 rad/s
Mode 1	7.874 Hz
Mode 2	19.570 Hz
ϕ_1	[0.768, 1.000]
ϕ_2	[0.433, -1.000]

The measured FFT value of 8.466 Hz is closest to the predicted first mode.

2-DOF RESULTS

The lower predicted mode is closest to the FFT statistic

The analytical model predicts both a lower in-phase mode and a higher out-of-phase mode.

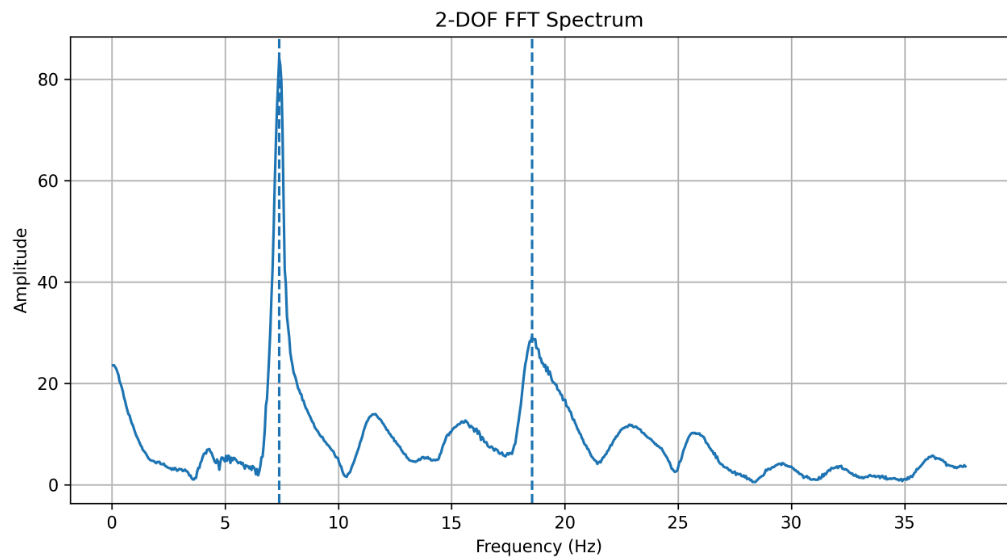
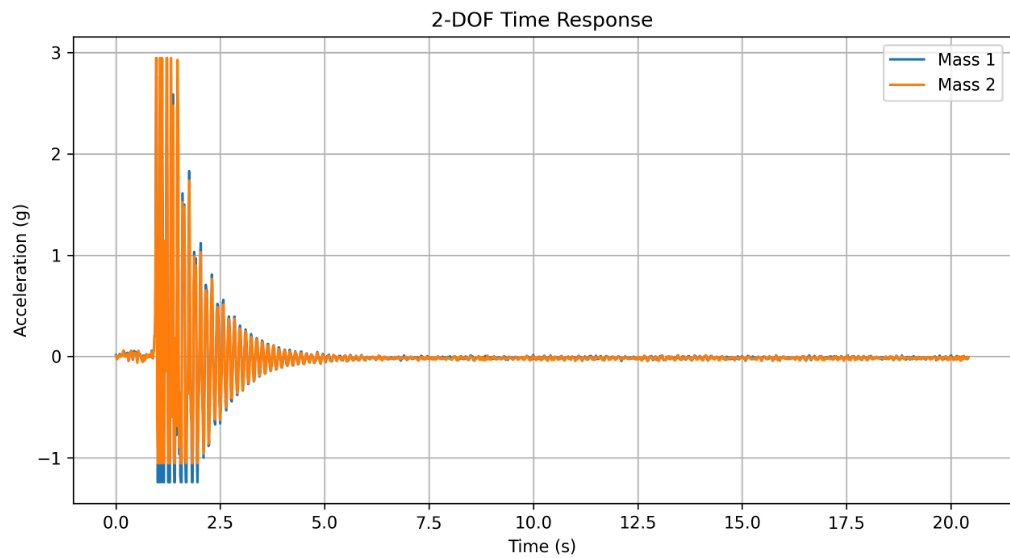


2-DOF SIGNAL EVIDENCE

The accelerometer signals show coupled motion



The time response shows both accelerometer channels, while the FFT spectrum separates the coupled signal into frequency content.



DISCUSSION AND CONCLUSION

The ideal model predicts the trend; the experiment reveals the real system

The measured differences are physically meaningful rather than just error. They point to damping, distributed mass, sensor mass, assembly compliance, and measurement-method sensitivity.

1-DOF conclusion

The theoretical frequency is 9.425 Hz, but peak counting and FFT give different measured values. That means the real oscillator is not perfectly ideal.

2-DOF conclusion

The mass and stiffness matrices predict two modes. The measured FFT value aligns most closely with the lower theoretical mode.

Hardware conclusion

Arduino accelerometer data makes the theory testable, while the physical printed spring and mounted electronics explain the deviations.

Theory gives the model; experiment tells us how the real system behaves.